

A note on the fractional-order Chua's system

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Abstract

This paper deals with fractional-order Chua's system. It is based on the well-known concept of the Chua's oscillator, where mathematical model of the Chua's system consists the fractional-order derivatives. We present the method to derive such kind of the fractional-order model. The fractional-order Chua's system with a total order less than three, which exhibits chaos as well as other nonlinear behavior, is presented. Numerical experimental example and real measurement are shown to verify the theoretical results.

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1. Introduction

There is an increase in the number of applications where fractional calculus has been used. This mathematical phenomenon allows to describe a real object more accurately than the classical “integer” methods. The real objects are generally fractional [22,19,21,16], however, for many of them the fractionality is very low. A typical example of a non-integer (fractional) order system is the voltage–current relation of a semi-infinite lossy transmission line [15] or diffusion of the heat into a semi-infinite solid, where heat flow is equal to the half-derivative of the temperature [19].

The main reason for using the integer-order models was the absence of solution methods for fractional differential equations. We have to identify and describe the real object by the fractional-order models. The first advantage is that we have more degrees of freedom in the model. The second advantage is that a “memory” is included in the model. Fractional-order systems have an unlimited memory, being integer-order systems cases in which the memory is limited.

It is well known that chaos cannot occur in continuous systems of total order less than three. This assertion is based on the usual concepts of order, such as the number of states in a system or the total number of separate differentiations or integrations in the system. The model of system can be rearranged to three single differential equations, where the equations contain the non-integer (fractional) order derivative. The total order of system is changed from 3 to the sum of each particular order. To put this fact into context, we can consider the fractional-order dynamical model of the system. Hartley et al. introduced the fractional-order Chua's system [17], in [1,30] the fractional-order cellular neural

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network was considered, in [7] the fractional Duffing's systems was presented, the other fractional-order chaotic systems were described in many others works (e.g. [3,4,6,5,9–12]). In all these cases, the chaos was exhibited in a system with total order less than three. From this consideration, the idea of developing a new fractional-order Chua's system arose. The term “system order” should be mentioned as well. The system order is not equal to the number of differential equations if we consider the fractional differential equations. The system order is equal to a highest derivative of the fractional differential equation of the mathematical model.

Arena and Hartley just simply replaced the fractional-order derivative instead of the integer order one. For numerical simulation, they used an approximation method proposed by Charef et al. [27]. This approximation of fractional-order operators is in the form of rational polynomial of high order in the frequency domain.

These considerations in the mentioned Arena's and Hartley's works lead to several notes. We will discuss three of them. The first note is on fractional-order of derivatives. We cannot just replace the order of derivatives from integer to fractional one without real reason. The reason is described in this paper. The second note is on used approximation methods. If we use a high order approximation method, then the total order of system is not equal to the highest derivative of the fractional differential equation but it is equal to the highest order of approximation polynomial. The third note is on system order. In both cases (Arena et al. and Hartley et al.), the terms of system order, model order, number of initial conditions, number of state-space variables and method for rewriting the state-space representation to fractional differential equation are not clearly described.

All mentioned notes will be explained in this paper, which is organized as follows. In Section 2, we briefly introduce the fractional calculus. Section 3 is on the basic concepts of Chua's system. The illustrative example is described in Section 4. Section 5 concludes this paper with some additional remarks.

2. Fractional calculus

2.1. Definitions of fractional derivatives

The idea of fractional calculus has been known since the development of the regular calculus, with the first reference probably being associated with Leibniz and L'Hospital in 1695 where half-order derivative was mentioned.

Fractional calculus is a generalization of integration and differentiation to non-integer order fundamental operator ${}_a D_t^r$, where a and t are the limits of the operation. The continuous integro-differential operator is defined as

$${}_a D_t^r = \begin{cases} \frac{d^r}{dt^r}, & \Re(r) > 0, \\ 1, & \Re(r) = 0, \\ \int_a^t (d\tau)^{-r}, & \Re(r) < 0. \end{cases}$$

The two definitions used for the general fractional differintegral are the Grünwald–Letnikov (GL) definition and the Riemann–Liouville (RL) definition [18,19]. The GL is given here as

$${}_a D_t^r f(t) = \lim_{h \rightarrow 0} h^{-r} \sum_{j=0}^{\left[\frac{t-a}{h}\right]} (-1)^j \binom{r}{j} f(t - jh), \quad (1)$$

where $[\cdot]$ means the integer part. The RL definition is given as

$${}_a D_t^r f(t) = \frac{1}{\Gamma(n-r)} \frac{d^n}{dt^n} \int_a^t \frac{f(\tau)}{(t-\tau)^{r-n+1}} d\tau \quad (2)$$

for $(n-1 < r < n)$ and where $\Gamma(\cdot)$ is the *Gamma* function.

2.2. Some properties of fractional derivatives

Two general properties of the fractional-order derivative will be used. The first is composition of fractional with integer-order derivative and the second is the property of linearity.

Similar to integer-order differentiation, fractional differentiation is a linear operation [19]:

$${}_a D_t^r (\lambda f(t) + \mu g(t)) = \lambda {}_a D_t^r f(t) + \mu {}_a D_t^r g(t). \quad (3)$$

The Laplace transform method is used for solving engineering problems. The formula for the Laplace transform of the RL fractional derivative (2) has the form [19]

$$\int_0^{\infty} e^{-st} {}_0D_t^r f(t) dt = s^r F(s) - \sum_{k=0}^{n-1} s^k {}_0D_t^{r-k-1} f(t) \Big|_{t=0}$$

for $(n-1 < r \leq n)$, where $s \equiv j\omega$ denotes the Laplace operator.

Some other important properties of the fractional derivatives and integrals can be found in several works (e.g. [18,19,22]).

2.3. Numerical methods for calculation of fractional derivatives

For numerical calculation of fractional-order derivation, we can use relation (4) derived from the Grunwald–Letnikov definition (1). This approach is based on the fact that for a wide class of functions, two definitions – GL (1) and RL (2) – are equivalent. The relation for the explicit numerical approximation of r th derivative at the points kT , ($k = 1, 2, \dots$) has the following form [19,23,26]:

$$({}_{(k-L_m/T)}D_{kT}^r f(t) \approx T^{-r} \sum_{j=0}^k (-1)^j \binom{r}{j} f_{k-j}, \quad (4)$$

where L_m is the “memory length”, T is the time step size of the calculation and $(-1)^j \binom{r}{j}$ are binomial coefficients $c_j^{(r)}$, ($j = 0, 1, \dots$). For their calculation, we can use the following expression [26]:

$$c_0^{(r)} = 1, \quad c_j^{(r)} = \left(1 - \frac{1+r}{j}\right) c_{j-1}^{(r)}. \quad (5)$$

The described numerical method is the so called Power Series Expansion (PSE) of a generating function. It is important to note that PSE leads to approximation in the form of polynomials, that is, the discretized fractional operator is in the form of FIR filter, which has only zeros. The other approach can be realized by Continued Fraction Expansion (CFE) of the generating function and then the discretized fractional operator is in the form of IIR filter, which has poles and zeros. In other words, for evaluation purposes, the rational approximations obtained by CFE frequently converge much more rapidly than the PSE and have a wider domain of convergence in the complex plane. On the other hand, the approximation by PSE and the short memory principle is convenient for the consideration of dynamical properties.

A detailed review of the other approximation methods and techniques (Carlson’s, Chareff’s, CFE, Matsuda’s, Oustaloup’s, etc.) for continuous and discrete fractional-order models in the form of IIR and FIR filters was done in [20,23]. Described forms of approximation were also compared with PSE and the others as for example Muir recursion [24]. Some other approaches were described in [14,20,22]. Last but not the least we should mention the approach proposed by Hwang, which is based on B-splines function [25].

2.4. Fractional calculus in electricity and magnetism

There are a large number of electric and magnetic phenomena where the fractional calculus can be used [16]. We will consider just two of them – capacitor and inductance coil behaviors.

Westerlund in 1994 proposed a new linear capacitor model [13]. It is based on Curie’s empirical law of 1889 which states that the current through a capacitor is

$$I(t) = \frac{V_0}{h_1 t^m},$$

where h_1 and m are constant, V_0 is the dc voltage applied at $t = 0$, and $0 < m < 1$.

For a general input voltage $V(t)$ the current is

$$I(t) = C_\phi \frac{d^m V(t)}{dt^m} \equiv C_{\phi 0} D_t^m V(t), \quad (6)$$

where C_ϕ is capacitance of the capacitor. It is related to the kind of dielectric. Another constant m (order) is related to the losses of the capacitor. Westerlund provided in his work the table of various capacitor dielectric with appropriated constant m which has been obtained experimentally by measurements.

Westerlund in his work also described the behavior of real inductor [16]. For a general current in the inductor, the voltage is

$$V(t) = L_\phi \frac{d^m I(t)}{dt^m} \equiv L_{\phi 0} D_t^m I(t), \quad (7)$$

where L_ϕ is inductance of the inductor and constant m is related to the “proximity effect”.

3. The concept of Chua's system

3.1. Classical Chua's oscillator

Classical Chua's oscillator, which is shown in Fig. 1(a), is a simple electronic circuit that exhibits nonlinear dynamical phenomena such as bifurcation and chaos. This circuit can be described by equations [31]:

$$\begin{aligned}\frac{dV_1(t)}{dt} &= \frac{1}{C_1} [G(V_2(t) - V_1(t)) - f(V_1(t))], \\ \frac{dV_2(t)}{dt} &= \frac{1}{C_2} [G(V_1(t) - V_2(t)) + I(t)], \\ \frac{dI(t)}{dt} &= \frac{1}{L_1} [-V_2(t) - R_L I(t)],\end{aligned}\quad (8)$$

where $G = 1/R_2$, $I(t)$ is the current through the inductance L_1 , $V_1(t)$ and $V_2(t)$ are the voltages over the capacitors C_1 and C_2 , respectively, and $f(V_1(t))$ is the piecewise-linear v - i characteristic of nonlinear resistor (NR) – Chua's diode, depicted in Fig. 1(b), which can be described as

$$I_{NR}(t) = f(V_1(t)) = G_b V_1(t) + \frac{1}{2} (G_a - G_b) (|V_1(t) + B_p| - |V_1(t) - B_p|) \quad (9)$$

with B_p being the breakpoint voltage of a diode, and $G_a < 0$ and $G_b < 0$ being some appropriate constants (slope of the piecewise-linear resistance).

By defining

$$\begin{aligned}x &= V_1/B_p, \quad y = V_2/B_p, \quad z = I_L/B_p G, \quad \alpha = C_2/C_1, \quad \beta = C_2/(L_1 G^2), \quad \gamma = C_2 R/(L_1 G), \quad m_1 = G_b/G, \\ m_0 &= G_a/G, \quad \tau = tG/C_2,\end{aligned}\quad (10)$$

we can transform (8) into the following corresponding dimensionless form of Chua's circuit [32]:

$$\begin{aligned}\frac{dx(t)}{dt} &= \alpha(y(t) - x(t) - f(x)), \\ \frac{dy(t)}{dt} &= x(t) - y(t) + z(t), \\ \frac{dz(t)}{dt} &= -\beta y(t) - \gamma z(t),\end{aligned}\quad (11)$$

where

$$f(x) = m_1 x(t) + \frac{1}{2} (m_0 - m_1) \times (|x(t) + 1| - |x(t) - 1|) \quad (12)$$

and τ in transformation equations (10) is the dimensionless time.

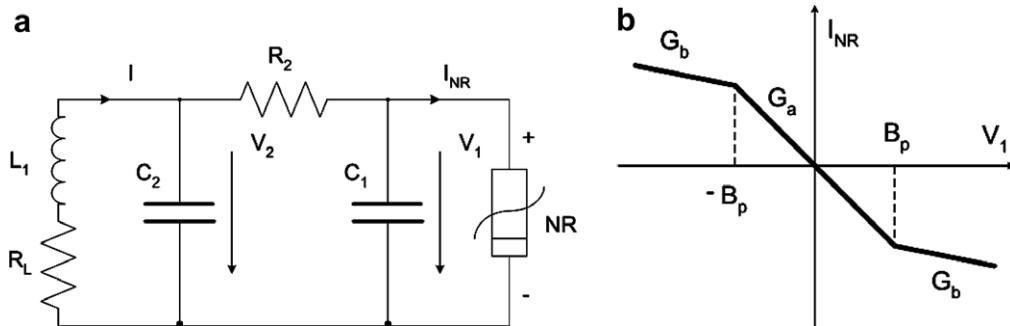


Fig. 1. Electrical circuit of Chua's oscillator and v - i characteristic of the nonlinear resistor. (a) Chua's circuit, (b) piecewise-linear characteristic.

3.2. Chua–Hartley’s oscillator

Given the techniques of fractional calculus, there are still a number of ways in which the order of system could be amended.

The Chua–Hartley’s system is different from the usual Chua’s system in that the piecewise-linear nonlinearity is replaced by an appropriate cubic nonlinearity which yields a very similar behavior. Derivatives on the left side of the differential equations are also replaced by the fractional derivatives as follows [17]:

$$\begin{aligned} {}_0D_t^q x(t) &= \alpha \left(y(t) + \frac{x(t) - 2x^3(t)}{7} \right), \\ {}_0D_t^q y(t) &= x(t) - y(t) + z(t), \\ {}_0D_t^q z(t) &= -\beta y(t) = -\frac{100}{7} y(t), \end{aligned} \quad (13)$$

where $q \leq 1$, $q \in \mathfrak{R}$ is the fractional-order of derivatives.

3.3. Chua–Podlubny’s oscillator

This system uses an approach where the order of any of three constitutive equations (8) can be changed so that the total order gives the desired value. In Chua–Podlubny’s case, in the equation 1 replaces the first differentiation by fractional differentiation of order $q < 1$, $q \in \mathfrak{R}$. The final dimensionless equations of the system is [19]

$$\begin{aligned} {}_0D_t^q x(t) &= \alpha {}_0D_t^{q-1} (y(t) - x(t)) - \frac{2\alpha}{7} (4x(t) - x^3(t)), \\ \frac{dy(t)}{dt} &= x(t) - y(t) + z(t), \\ \frac{dz(t)}{dt} &= -\frac{100}{7} y(t) = -\beta y(t), \end{aligned} \quad (14)$$

where $\alpha = C_2/C_1$ and $\beta = C_2 R_2^2/L_1$.

A similar system was proposed in [28] but instead the fractional derivative on the left side of Eqs. (14) was used as an integer order one.

3.4. New fractional-order Chua’s oscillator

The circuit behavior can be described by three fractional differential equations with the various orders. Applying the Kirchhoff laws for two current nodes and one voltage loop and relations (6) and (7) into circuit depicted in Fig. 1(a), we obtain the following mathematical model of circuit for state variables $V_1(t)$, $V_2(t)$ and $I(t)$:

$$\begin{aligned} C_1 {}_0D_t^{q_1} V_1(t) + I_{NR}(t) &= \frac{V_2(t) - V_1(t)}{R_2}, \\ C_2 {}_0D_t^{q_2} V_2(t) - I(t) &= \frac{V_1(t) - V_2(t)}{R_2}, \\ L_1 {}_0D_t^{q_3} I(t) + V_2(t) + R_L I(t) &= 0. \end{aligned} \quad (15)$$

Eqs. (15) can be rewritten in the following form [29]:

$$\begin{aligned} {}_0D_t^{q_1} V_1(t) &= \frac{1}{C_1 R_2} [V_2(t) - V_1(t)] - \frac{f(V_1(t))}{C_1}, \\ {}_0D_t^{q_2} V_2(t) &= \frac{1}{C_2 R_2} [V_1(t) - V_2(t)] + \frac{I(t)}{C_2}, \\ {}_0D_t^{q_3} I(t) &= \frac{1}{L_1} [-V_2(t) - R_L I(t)], \end{aligned} \quad (16)$$

where V_1 is a voltage over the capacitor C_1 , V_2 is a voltage over the capacitor C_2 , I is a current through the inductance L_1 , q_1 is a real order of the capacitor C_1 , q_2 is a real order of the capacitor C_2 , q_3 is a real order of the inductor L_1 , $f(V_1)$ is a piecewise-linear v – i characteristic of nonlinear Chua’s diode, which can be described by (9).

By using transformation (10), we can rewrite Eqs. (16) in the following dimensionless form:

$$\begin{aligned}
{}_0D_t^{q_1}x(t) &= \alpha(y(t) - x(t) - f(x)), \\
{}_0D_t^{q_2}y(t) &= x(t) - y(t) + z(t), \\
{}_0D_t^{q_3}z(t) &= -\beta y(t) - \gamma z(t),
\end{aligned} \tag{17}$$

where $f(x)$ is the piecewise-linear nonlinearity (12).

4. Illustrative example

4.1. Experimental measurements

Classical Chua's oscillator can be realized by electrical elements according to the scheme shown in Fig. 2, which is a simple electronic circuit [8] that exhibits nonlinear dynamical phenomena such as bifurcation and chaos. Chua's diode (9) – negative impedance converter – nonlinear resistor – was realized by operating amplifier LM 358 and resistors R_1 , R_7 and R_8 ($R_7 = R_8$).

For experimental verification of Chua's system depicted in Fig. 2 and described by Eqs. (16) and (9), the following values of electrical elements were chosen:

$$\begin{aligned}
C_1 &= 4.71 \text{ nF}, \quad C_2 = 48 \text{ nF}, \quad L_1 = 4.64 \text{ mH}, \\
R_L &= 15.8 \, \Omega, \quad R_1 = 897 \, \Omega, \quad R_2 = 998 \, \Omega, \quad R_7 = R_8 = 393 \, \Omega.
\end{aligned} \tag{18}$$

We used the metalized paper capacitors C_1 and C_2 with the real order $q_1 = q_2 = 0.98$ and we assume the real order of inductor to be $q_3 = 0.94$ [16]. Total order of the system is $\bar{q} = 2.90$.

The measured breakpoints of the nonlinear characteristic (9) are

$$-B_p = (-8.79 \text{ V}, 7.7 \text{ mA}), \quad B_p = (9.12 \text{ V}, -7.9 \text{ mA}).$$

Assuming the three-segment piecewise-linear voltage–current transfer characteristic of negative impedance converter (9), we have the slope $G_a = -1/R_1 = -1.1148 \text{ mA/V}$ for $R_7 = R_8$ and the slope G_b was calculated using the breakpoints B_p and it has the value $G_b = -0.8710 \text{ mA/V}$.

The resistors R_3 , R_4 , R_5 , R_6 , and the diodes D_1 and D_2 generate the positive and negative half of nonlinearity.

In Fig. 3(a), the photo of the digital oscilloscope screen (Tektronix TDS1002, 60 Mhz) is depicted. It is a real measurement of voltages V_1 – V_2 for circuit presented in Fig. 2 with the parameters of electrical components (18). The result shown in Fig. 3(a) is the double-scroll attractor of fractional-order Chua's system described by Eqs. (16) and (9).

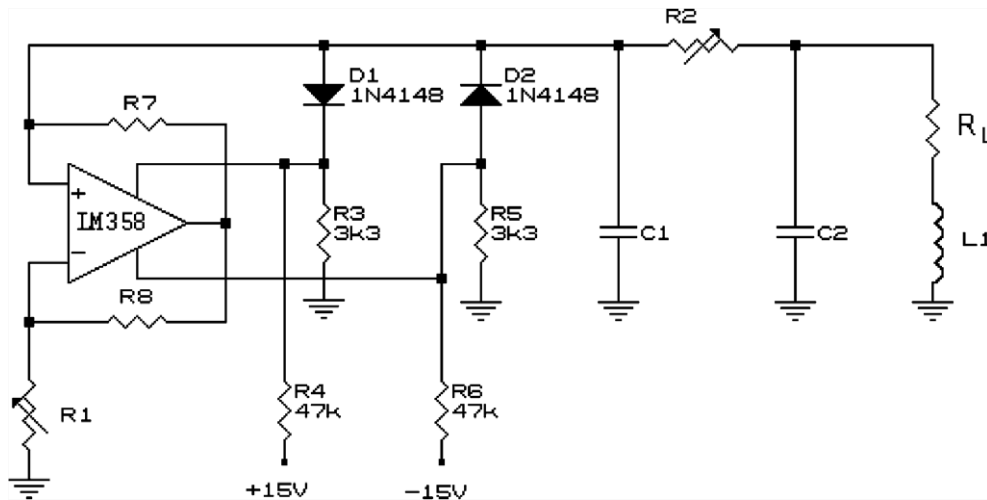


Fig. 2. Practical realization of Chua's circuit.

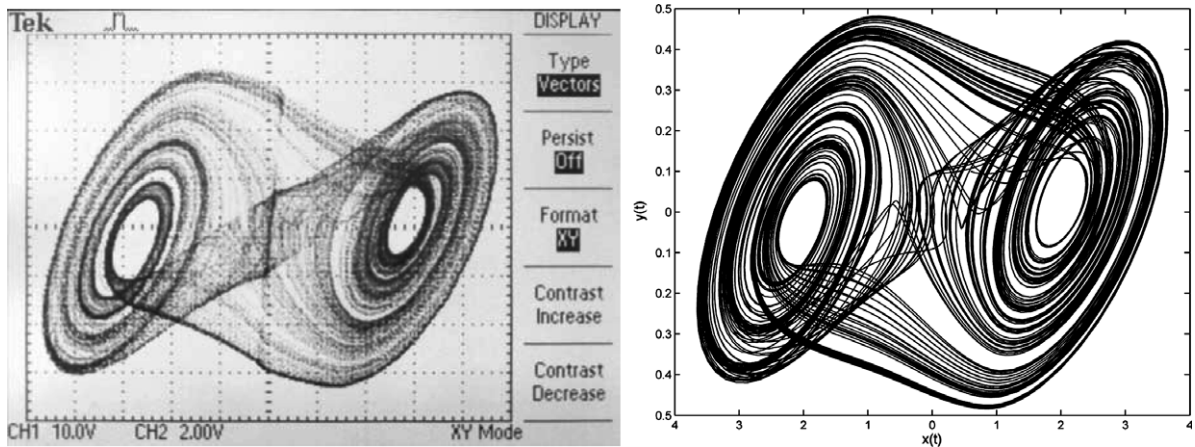


Fig. 3. Strange attractors of the Chua's system in 2D. (a) Oscilloscope screen: measured double-scroll attractor of the Chua's system (16), (b) simulation result: double-scroll attractor of the Chua's system (17).

4.2. Simulation results

The similar and comparable results, the fractional-order Chua's system behavior, can be obtained by numerical simulation with utilization of Eq. (4) for time step $T = 0.001$ [s], $L_m = 10$ (10,000 values and coefficients from history). For the computation of the fractional-order derivative in Eqs. (16), relations (4), (5) and properties (3) can be used.

Fig. 3(b) shows the double-scroll attractor of Chua's circuit (17) with total order $\bar{q} = 2.90$ for the parameters: $\alpha = 10.1911$, $\beta = 10.3035$, $\gamma = 0.1631$, $q_1 = q_2 = 0.98$, $q_3 = 0.94$, $m_0 = -1.1126$ and $m_1 = -0.8692$ computed numerically for initial conditions $(x(0), y(0), z(0)) = (0.6, 0.1, -0.6)$ and the parts values (18).

5. Conclusions

In this paper, we have presented an approach to derive the fractional-order Chua's system and a method for its numerical simulation. By an illustrative example, we have shown the behavior of a new fractional-order Chua's system.

In this section, we will also discuss the second and the third notes mentioned in introduction subsection. Till the present time, no analytical solution has been found for chaotic systems. Some authors consider the numerical solutions (attractors) as a numerical error. In fact, the deterministic chaos exists but computation of strange attractors is a very important thing and therefore we have to find appropriate approximation methods. Utilization of methods in the form of rational polynomial leads to high order system. In this case, we must consider different initial conditions and large numerical error which is amplified by system constant and approximation polynomial constants. We recommend to use a method in the form of FIR filter with a large number of coefficients. It is also a high order system but numerical error is much smaller than in methods in the form of IIR filter [23,24]. However, the time of computation is longer because of the number of coefficients.

System order in such a case is equal to the sum of particular fractional orders of differential equations. The conclusion of this work confirms the conclusions of [2,17,19] that there is a need to refine the notion of the order of a system which cannot be considered only by the total number of differentiations. For fractional-order differential equations, the number of terms is more important than the order of differentiation.

We have considered an example of chaotic fractional-order Chua's circuit, which exhibits chaotic behavior with total order less than three. As has been demonstrated, the idea of fractional calculus requires one to reconsider dynamic system concepts that are often taken for granted. So by changing the order of a system from integer to real, we also move from a three-dimensional system to infinite dimension.

The fractional-order model for chaotic Chua's system was directly derived because electrical elements used in circuit are not ideal. As was mentioned in [16], the real electrical elements (e.g. capacitors, inductors) have fractional-order and should be described by the fractional-order models.

As has been demonstrated, the idea of fractional calculus requires reconsideration of dynamical system concepts. Some of them have been noted in this article.

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